Questions

Q1.

In this question you must show detailed reasoning.

Solutions relying on calculator technology are not acceptable.

$$f(x) = x^2(2x + 1) - 15x$$

(a) Solve

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$$f(x) = 0$$

(4)

(b) Hence solve

$$y^{\frac{4}{3}}\left(2y^{\frac{2}{3}} + 1\right) - 15y^{\frac{2}{3}} = 0 \qquad y > 0$$

giving your answer in simplified surd form.

(2)

(Total for question = 6 marks)

(**Q04 WMA11/01, Oct 2022**)

Q2.

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Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form $a + b\sqrt{c}$, where a, b and c are integers to be found.

(5)

(Total for question = 5 marks)

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(Q08 WMA11/01, Oct 2019)

Q3.

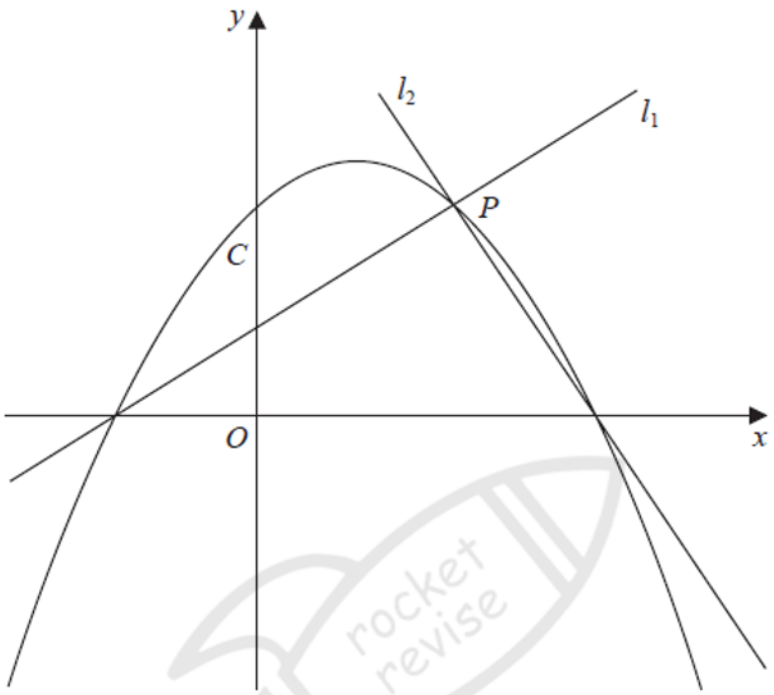


Figure 5

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x + 2)(x - b)$$ where b is a positive constant

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$
- intersects C on the negative x-axis and at the point P

(a) (i) Write down an equation for l_1

(1)

(ii) Find, in terms of b, the coordinates of P

(3)

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x-axis,

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(b) find, in terms of b, an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

(d) Hence, or otherwise, find the value of b

(2)

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(Total for question = 10 marks)

(Q10 WMA11/01, June 2023)

Q4.

$$f(x) = 11 - 4x - 2x^2$$

(a) Express $f(x)$ in the form

$$a + b(x + c)^2$$

where a, b and c are integers to be found.

(3)

(b) Sketch the graph of the curve C with equation $y = f(x)$, showing clearly the coordinates of the point where the curve crosses the y-axis.

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(2)

(c) Write down the equation of the line of symmetry of C.

(1)

(Total for question = 6 marks)

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(Q02 WMA11/01, Jan 2022)

Q5.

(a) Express $3x^2 + 12x + 13$ in the form

$$a(x + b)^2 + c$$

where a, b and c are integers to be found.

(3)

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(b) Hence sketch the curve with equation $y = 3x^2 + 12x + 13$

On your sketch show clearly

- the coordinates of the y intercept
- the coordinates of the turning point of the curve

(3)

(Total for question = 6 marks)

(Q03 WMA11/01, June 2023)

Q6.

(a) Find, using algebra, all real solutions of

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$$2x^3 + 3x^2 - 35x = 0$$

(3)

(b) Hence find all real solutions of

$$2(y - 5)^6 + 3(y - 5)^4 - 35(y - 5)^2 = 0$$

(4)

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(Total for question = 7 marks)

(Q05 WMA11/01, June 2019)

Q7.

(a) Find, using algebra, all solutions of

$$20x^3 - 50x^2 - 30x = 0$$

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(3)

(b) Hence find all real solutions of

$$20(y+3)^{\frac{3}{2}} - 50(y+3) - 30(y+3)^{\frac{1}{2}} = 0$$

(4)

(Total for question = 7 marks)

(Q05 WMA11/01, Jan 2020)

Q8.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

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$$f(x) = ax^3 + (6a + 8)x^2 - a^2x$$

where a is a positive constant.

Given $f(-1) = 32$

(a) (i) show that the only possible value for a is 3

(ii) Using $a = 3$ solve the equation

$$f(x) = 0$$

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(5)

(b) Hence find all real solutions of

(i) $3y + 26y^{\frac{2}{3}} - 9y^{\frac{1}{3}} = 0$

(ii) $3\left(9^{3z}\right) + 26\left(9^{2z}\right) - 9\left(9^{z}\right) = 0$

(5)

(Total for question = 10 marks)

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(Q02 WMA11/01, June 2021)

Q9.

The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

(a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a, b and c are constants to be found.

(3)

The point P is the minimum point of C_1

(b) Deduce the coordinates of P.

(1)

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A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A, B, C and D are constants.

Given that C_2

- passes through P
- intersects the x-axis at –4, –2 and 3

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(c) find, making your method clear, the values of A, B, C and D.

(5)

(Total for question = 9 marks)

(Q08 WMA11/01, June 2021)

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Q10.

$$f(x) = 3 + 12x - 2x^2$$

(a) Express $f(x)$ in the form

$$a - b(x + c)^2$$

where a, b and c are integers to be found.

(3)

The curve with equation $y = f(x) - 7$ crosses the x-axis at the points P and Q and crosses the y-axis at the point R.

(b) Find the area of the triangle PQR, giving your answer in the form $m\sqrt{n}$ where m and n are integers to be found.

(4)

(Total for question = 7 marks)

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(Q02 WMA11/01, Oct 2020)

Q11.

The line with equation $y = 4x + c$, where c is a constant, meets the curve with equation $y = x(x - 3)$ at only one point.

(a) Find the value of c.

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(4)

(b) Hence find the coordinates of the point of intersection.

(3)

(Total for question = 7 marks)

(Q06 WMA11/01, June 2019)

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